Handoff Scenario in a Mobile Network

A handoff occurs when a mobile device moves from one cell to another while maintaining an active connection.

1. Mobile Moves to New Cell

- The mobile device moves from Cell A to Cell B.
- Signal strength from Cell B increases while Cell A's signal decreases.

2. Measurement & Decision

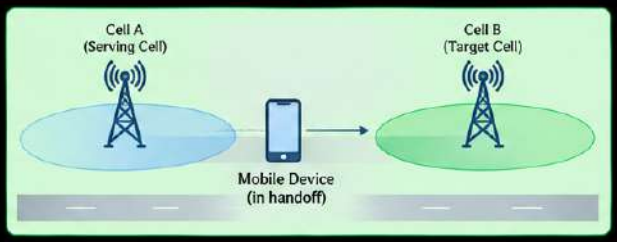
- The mobile monitors signal quality (RSSI/RSRP).
- The network (via BSC/eNB) decides to initiate a handoff.
- Handoff preparation begins.

3. Handoff Request

- The mobile sends a handoff request to the serving base station (Cell A).
- The base station forwards the request to the target base station (Cell B).

4. Resource Allocation

- Target cell (Cell B) reserves radio resources.
- The network updates the routing information (MSC/VLR).
- A synchronization signal is sent to the mobile.



5. Handoff Execution

- The mobile switches its connection from Cell A to Cell B.
- Data flow is redirected to the new cell.
- The old channel is released.

6. Update & Confirmation

- The network updates the location information.
- A handoff complete message is sent to the mobile.
- The connection is now active in Cell B.

7. Resume Data Transfer

- The mobile resumes data transfer using the new cell's resources.
- Voice, data, and other services continue without interruption (or with minimal delay).

8. Ongoing Monitoring

- The network continues to monitor signal strength and quality.
- Further handoffs may be triggered as the mobile moves.